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Lab-1

Sigmoid derivative.

$$g(z) = \frac{1}{1+e^{-z}}$$

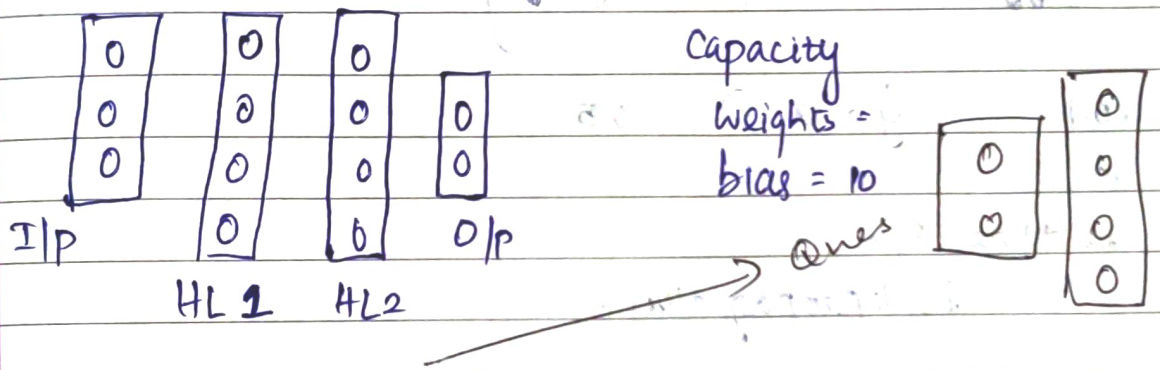
$$g(z') = \frac{1}{1+e^{-z}} \times \frac{1+e^{-z}}{1+e^{-z}}$$

$$\frac{d}{dz} g'(z) = d/dz \, (1+e^{-z})^{-1}$$

$$= -1 \, (1+e^{-z})^{-2}$$

21.07.2026 **Lecture-4**

- Feed Forward Network.



$W_{11}^{[1]} = 0.1$ $W_{21}^{[1]} = -0.1$ $W_{31}^{[1]} = 0.2$ $W_{12}^{[1]} = -1.1$ $W_{22}^{[1]} = 0.4$ $W_{32}^{[1]} = 1.1$

$W_{11}^{[2]} = 0.2$ $W_{21}^{[2]} = 0.3$ $W_{31}^{[2]} = 0.1$ $W_{12}^{[2]} = -0.1$ $W_{22}^{[2]} = -0.2$ $W_{32}^{[2]} = -0.1$

$W_{11}^{[3]} = 0.3$ $W_{12}^{[3]} = -0.3$

Activation function

$a = g(z) = \frac{1}{1+e^{-z}}$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.3 \\ -1.2 \end{bmatrix}$$

$$[(0.1)(0.3) + (-1.1)(+0.3)]$$

$$= 0.03 + 0.33 = 0.36 = z.$$

$$\frac{1}{1+e^{-(0.36)}} = \boxed{0.589}$$

11 x 03
33
00
0.33

$= [-0.1 \times 0.3] + [0.4 \times 0.3]$

$= -0.03 + 0.12$

$= 0.09$

CHECK THIS

$Z_1^{[1]} = 1.35$	$a_1^{[1]} = 0.79$
$Z_2^{[1]} = -0.5$	$a_2^{[1]} = 0.37$
$Z_3^{[1]} = -1.26$	$a_3^{[1]} = 0.22$

$Z_1^{[2]} = 0.15$	$a_1^{[2]} = 0.53$
$Z_2^{[2]} = 0.17$	$a_2^{[2]} = 0.54$

$Z_1^{[3]} = 0.00$	$a_1^{[3]} = 0.5$

~~[0.2 x 0.1~~ $\hat{y} = a_1^{[3]} = 0.5$

22.07.2026 Lecture -5,6.

- Backpropagation.

Learning:-

1) Forward Pass.
2) Compute the error or loss
3) $\theta^{new} = \theta^{old} - \alpha \nabla J$

$\theta_1 = \theta_1 - \alpha \, \partial J / \partial \theta$

$h_\theta x = \theta^T x$

$J = 1/2 (h_\theta(x) - y)^2$

$\partial J / \partial \theta_1 = (h_\theta(x) - y) x$